The Lorentz-Invariant Classical Wave Equation for a Massless Scalar Field

A Four-Dimensional Formulation in Minkowski Spacetime

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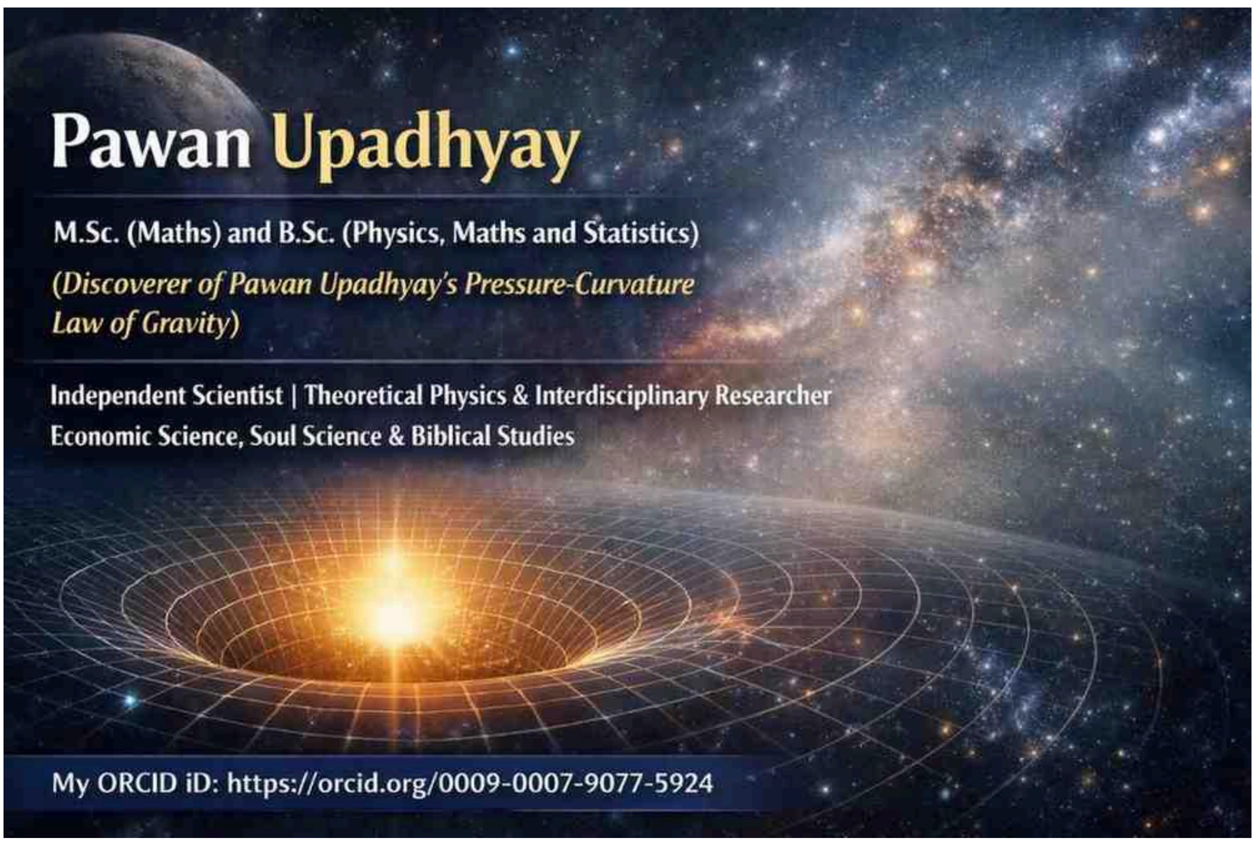

Abstract

The classical wave equation is traditionally written as

$$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi$$

and is commonly associated with the propagation of disturbances through a physical medium. However, the mathematical structure of the wave equation is more general than mechanical wave motion. When formulated in four-dimensional spacetime using the coordinates **(x, y, z, ct)**, the equation takes the Lorentz-invariant form $\Box\varphi = 0$. This equation describes the propagation of a free massless scalar field in Minkowski spacetime.

This paper examines the mathematical transition from the conventional three-dimensional wave equation to its four-dimensional relativistic form. The spacetime metric, d'Alembertian operator, plane-wave solutions, dispersion relation, null propagation, and massless character of the field are developed. The distinction between mechanical waves and classical field waves is also examined. Finally, the scalar wave equation is compared with the electromagnetic wave equations derived from Maxwell's theory.

Keywords: classical wave equation, scalar field, Lorentz invariance, Minkowski spacetime, d'Alembertian, massless field, four-dimensional spacetime, wave propagation.

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1. Introduction

The wave equation occupies a central position in classical mathematical physics. It appears in the theory of vibrating strings, acoustics, elasticity, electromagnetism, and many other areas of physics.

In its familiar form, the scalar wave equation is

$$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi$$

where φ represents the quantity undergoing wave propagation and c represents the characteristic propagation velocity.

In mechanical systems, φ may represent a displacement, pressure variation, or another physical quantity associated with matter. Such examples naturally lead to the idea that a wave is a disturbance of a material medium.

The development of relativistic physics provides a broader perspective. A field can possess wave dynamics without representing the mechanical displacement of a material substance. In particular, the electromagnetic field propagates through vacuum and is described classically by Maxwell's equations.

The purpose of this paper is to investigate the four-dimensional formulation of the classical wave equation using the spacetime coordinates

(x, y, z, ct).

The central equation considered is

$$\square\varphi = 0$$

where $\square$ denotes the d'Alembertian operator.

The objective is not to replace the ordinary wave equation but to demonstrate that its relativistic structure is naturally expressed through a four-dimensional spacetime operator.

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2. The Classical Wave Equation

Let

$$\varphi = \varphi(x, y, z, t)$$

be a scalar field.

The three-dimensional Laplacian is

$$\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2.$$

The classical wave equation is therefore

$$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi.$$

Expanding the Laplacian gives

$$\partial^2\varphi/\partial t^2 = c^2(\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2).$$

Dividing by c^2 and rearranging,

$$\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2 - (1/c^2)\partial^2\varphi/\partial t^2 = 0.$$

This equation already contains the mathematical relationship between spatial variation and temporal variation that characterizes wave propagation.

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3. Four-Dimensional Spacetime Coordinates

We now introduce the four coordinates

$$x^u = (x, y, z, ct).$$

The field becomes

$$\varphi = \varphi(x, y, z, ct).$$

The use of ct rather than t is dimensionally convenient because x, y, z, and ct all have dimensions of length.

The fourth coordinate is nevertheless fundamentally different from the three spatial coordinates. It represents time multiplied by the invariant speed c.

The spacetime interval is

$$ds^2 = dx^2 + dy^2 + dz^2 - c^2dt^2.$$

Equivalently,

$$ds^2 = dx^2 + dy^2 + dz^2 - d(ct)^2.$$

The minus sign distinguishes Minkowski spacetime from ordinary Euclidean four-dimensional space.

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4. Minkowski Metric

Using the metric convention

$$g_{uv} = \mathrm{diag}(1, 1, 1, -1),$$

the spacetime interval can be written as

$$ds^2 = g_{uv}\, dx^u\, dx_v.$$

The inverse metric has the same diagonal form:

$$g^u{}_v = \mathrm{diag}(1, 1, 1, -1).$$

This metric determines how spatial and temporal derivatives combine to produce a Lorentz-invariant differential operator.

The essential feature is that the temporal component enters with the opposite sign from the spatial components.

5. The Four-Dimensional Wave Operator

Define the four-dimensional derivative

$$\partial_u = \partial/\partial x^u.$$

The Lorentz-invariant wave operator is defined by

$$\Box = g^u{}_v \partial_u \partial_v.$$

For the metric chosen above,

$$\Box = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2 - \partial^2/\partial(ct)^2.$$

Since

$$\partial/\partial(ct) = (1/c)\partial/\partial t,$$

we obtain

$$\partial^2/\partial(ct)^2 = (1/c^2)\partial^2/\partial t^2.$$

Therefore,

$$\Box = \nabla^2 - (1/c^2)\partial^2/\partial t^2.$$

6. The Lorentz-Invariant Classical Wave Equation

The source-free equation is

$$\Box\varphi = 0.$$

Expanding,

$$\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2 - \partial^2\varphi/\partial(ct)^2 = 0.$$

Equivalently,

$$\nabla^2\varphi - (1/c^2)\partial^2\varphi/\partial t^2 = 0.$$

Rearranging,

$$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi.$$

Thus, the conventional classical wave equation is recovered exactly.

The four-dimensional formulation therefore does not introduce a different wave equation. Instead, it reveals the spacetime structure of the equation.

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7. Lorentz Invariance

A Lorentz transformation connects the coordinates of two inertial observers:

$$x^u \rightarrow x'^u.$$

The defining property of the transformation is the invariance of the spacetime interval:

$$ds'^2 = ds^2.$$

For a scalar field,

$$\varphi'(x') = \varphi(x).$$

The d'Alembertian is constructed from the Minkowski metric and therefore has invariant form.

Consequently,

$$\Box\varphi = 0$$

has the same mathematical form in all inertial frames.

This property distinguishes the relativistic wave equation from a generic wave equation written in an arbitrary coordinate system.

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8. Plane-Wave Solution

Consider the plane-wave solution

$\varphi = A\, e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)}$

where

$\mathbf{r} = (x, y, z)$

and

$\mathbf{k} = (k_x, k_y, k_z)$.

For this field,

$\nabla^2\varphi = -k^2\varphi$

and

$\partial^2\varphi/\partial t^2 = -\omega^2\varphi$.

Substitution into the wave equation gives

$-k^2\varphi + (\omega^2/c^2)\varphi = 0$.

For a nonzero field,

$\omega^2/c^2 = k^2$.

Therefore,

$\omega^2 = c^2k^2$

and hence

$\omega = ck$.

This is the dispersion relation for the massless wave.

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9. The Four-Wave-Vector

Define the wave four-vector by

$k^\mu = (k_x, k_y, k_z, \omega/c)$.

The plane wave can be expressed in four-dimensional notation as

$\boldsymbol{\varphi = A\, e^{ik_u x^u}}$.

The wave equation requires

$\boldsymbol{k_u k^u = 0}$.

Using the adopted metric,

$\boldsymbol{k_u k^u = k^2 - \omega^2/c^2}$.

Therefore,

$\boldsymbol{k_u k^u = 0}$

implies

$\boldsymbol{\omega^2 = c^2 k^2}$.

The four-wave-vector is consequently a null vector.

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10. Massless Character

The relativistic scalar-field equation with mass m is, subject to metric convention,

$\boldsymbol{(\square + m^2c^2/\hbar^2)\varphi = 0}$.

When

$\boldsymbol{m = 0}$,

the mass term disappears and the equation becomes

$\boldsymbol{\square\varphi = 0}$.

Thus the source-free scalar wave equation is the massless limit of the relativistic scalar-field equation.

The corresponding dispersion relation is

$\boldsymbol{\omega^2 = c^2 k^2}$.

For a massive field, an additional mass-dependent term appears in the dispersion relation. The absence of such a term is characteristic of the massless case.

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11. Null Propagation and the Light Cone

The spacetime interval is

$$ds^2 = dx^2 + dy^2 + dz^2 - c^2dt^2.$$

For a null trajectory,

$$ds^2 = 0.$$

Therefore,

$$dx^2 + dy^2 + dz^2 = c^2dt^2.$$

Defining

$$d\ell^2 = dx^2 + dy^2 + dz^2,$$

we obtain

$$d\ell = c\, dt.$$

Consequently,

$$d\ell/dt = c.$$

The wave equation is therefore associated with propagation along the null structure of Minkowski spacetime.

The set of null directions forms the light cone.

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12. Classical Field Versus Mechanical Wave

A central conceptual issue is the distinction between a classical wave and a mechanical wave.

A mechanical wave requires a physical system whose mechanical degrees of freedom participate in the disturbance.

For example, a string displacement u may satisfy

$$\partial^2 u/\partial t^2 = v^2 \partial^2 u/\partial x^2.$$

Here u describes a mechanical displacement.

The scalar field φ in

$$\Box\varphi = 0$$

does not necessarily have such an interpretation.

It can instead represent a fundamental classical field.

Therefore, the mathematical existence of a wave equation does not by itself imply the existence of a material medium.

This distinction is important when discussing electromagnetic radiation and other field waves.

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13. Electromagnetic Waves

Electromagnetic radiation is described fundamentally by Maxwell's equations.

In vacuum, Maxwell's equations imply wave equations for the electric and magnetic fields:

$$\Box \mathbf{E} = 0$$

and

$$\Box \mathbf{B} = 0.$$

These equations have the same d'Alembertian structure as

$$\Box\varphi = 0.$$

However, the electromagnetic field is not a scalar field. It possesses vector and tensor structure.

The electromagnetic field can be represented covariantly by the antisymmetric field tensor

$$F_{uv}.$$

Therefore, the scalar equation should not be identified as the complete electromagnetic theory.

Rather, it demonstrates the general mathematical structure shared by massless relativistic waves.

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14. Lagrangian Formulation

The equation

$$\Box\varphi = 0$$

can be obtained from a Lorentz-invariant classical Lagrangian.

For a free real scalar field, consider

$$\mathcal{L} = \tfrac{1}{2}\, \partial_u\varphi\, \partial^u\varphi.$$

The action is

$$S = \int \mathcal{L}\, d^4x.$$

Applying the Euler-Lagrange field equation produces

$$\partial_u\partial^u\varphi = 0.$$

Therefore,

$$\Box\varphi = 0.$$

This result demonstrates that the wave equation can be formulated consistently within classical relativistic field theory.

The field does not have to be interpreted mechanically for the variational formulation to exist.

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15. Energy and Momentum

A classical scalar field can possess energy and momentum.

For the free scalar field, the energy-momentum tensor can be written, with the corresponding sign convention, as

$$T^u{}_v = \partial^u\varphi\, \partial^v\varphi - \tfrac{1}{2}g^u{}_v\, \partial_a\varphi\, \partial^a\varphi.$$

The field equation implies the conservation relation

$$\partial_u T^u{}_v = 0.$$

Thus, the field can transport conserved physical quantities even though it is not interpreted as a mechanical displacement of matter.

This provides an important distinction between the concepts of **field propagation** and **material motion**.

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16. Source-Free and Source-Driven Equations

The present paper considers the source-free equation

$$\Box\varphi = 0.$$

A source can be introduced through an equation of the form

$$\Box\varphi = J.$$

Here J represents a source term, with its precise normalization depending on the physical theory and convention.

The source-free equation describes free propagation, whereas the sourced equation describes a field influenced by an external or internal source.

This distinction is analogous to the difference between free electromagnetic radiation and electromagnetic fields generated by charges and currents.

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17. Why Four Dimensions Matter

The expression

$$\varphi = \varphi(x, y, z, ct)$$

makes explicit that the field is defined over spacetime.

The wave equation then becomes

$$\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2 - \partial^2\varphi/\partial(ct)^2 = 0.$$

The equation cannot be understood correctly by treating all four coordinates as equivalent Euclidean directions.

The temporal coordinate has a different metric sign.

Thus, the essential structure is not simply

four coordinates

but rather

four-dimensional Lorentzian spacetime.

This Lorentzian structure produces the causal light-cone geometry characteristic of relativistic wave propagation.

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18. Discussion

The four-dimensional formulation provides a deeper interpretation of the classical wave equation.

The ordinary form

$$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi$$

emphasizes the relationship between spatial curvature and temporal acceleration.

The relativistic form

$$\square\varphi = 0$$

emphasizes that the field exists on spacetime and that its differential equation is compatible with Lorentz symmetry.

The two descriptions are mathematically equivalent when the Minkowski spacetime structure is understood.

The four-dimensional form is therefore not merely a change of notation. It exposes the geometric principle underlying relativistic wave propagation.

The scalar field itself is not necessarily a material displacement. The field can be treated as a classical dynamical entity whose evolution is governed by the spacetime differential operator.

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19. Limitations

The equation

$$\Box\varphi = 0$$

should not be interpreted as a universal equation for all waves.

Different physical systems require different dynamical equations.

Mechanical waves depend on properties of matter such as density, elasticity, and pressure.

Electromagnetic waves require Maxwell's equations.

The scalar equation is therefore best interpreted specifically as the equation governing a free massless scalar field.

Its conceptual importance lies in demonstrating how wave propagation can be expressed independently of a mechanical medium.

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20. Conclusion

The classical wave equation

$$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi$$

can be naturally expressed in four-dimensional spacetime using the coordinates

$$(x, y, z, ct).$$

With the Minkowski metric

$$g_{uv} = \mathrm{diag}(1, 1, 1, -1),$$

the corresponding wave operator is

$$\Box = \nabla^2 - (1/c^2)\partial^2/\partial t^2.$$

The source-free equation is therefore

$$\Box\boldsymbol{\varphi} = 0.$$

Its plane-wave solution,

$$\boldsymbol{\varphi} = A\, e^{ik\cdot r - i}\omega t,$$

leads to the dispersion relation

$$\omega^2 = c^2k^2.$$

In four-vector language, this is expressed as

$$k_u k^u = 0,$$

showing that the wave four-vector is null.

The analysis establishes that the classical massless scalar wave equation is naturally a Lorentz-invariant equation on Minkowski spacetime. Its mathematical structure does not require a mechanical medium. Instead, it describes the propagation of a classical field whose physical interpretation must be supplied by the relevant field theory.

The central equation of the analysis is therefore:

$$\Box\boldsymbol{\varphi} = 0$$

or, explicitly,

$$\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2 - \partial^2\varphi/\partial(ct)^2 = 0.$$

Equivalently,

$$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi.$$

The significance of this formulation lies in the fact that the familiar classical wave equation can be understood as a manifestation of the geometry of spacetime and the dynamics of a field, rather than necessarily as the mechanical oscillation of a material medium.

Final Mathematical Summary

Spacetime coordinates:

$x^u = (x, y, z, ct)$

Spacetime interval:

$ds^2 = dx^2 + dy^2 + dz^2 - c^2dt^2$

Wave operator:

$\square = \nabla^2 - (1/c^2)\partial^2/\partial t^2$

Massless scalar-field equation:

$\square\varphi = 0$

Expanded equation:

$\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2 - \partial^2\varphi/\partial(ct)^2 = 0$

Equivalent classical form:

$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi$

Plane wave:

$\varphi = A\ e^{i(k\cdot r - \omega t)}$

Dispersion relation:

$\omega^2 = c^2k^2$

Null condition:

$k_u k^u = 0$

Massless limit:

$m = 0 \rightarrow \square\varphi = 0$

Central proposition:

A free massless classical scalar field propagating on Minkowski spacetime satisfies the Lorentz-invariant equation $\boldsymbol{\square\varphi = 0}$. :::

[In Greek Symbol]

The Lorentz-Invariant Classical Wave Equation for a Massless Scalar Field

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Abstract

The classical wave equation is traditionally written as

$$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi$$

and is widely used to describe the propagation of disturbances and fields. When formulated in four-dimensional spacetime using the coordinates **(x, y, z, ct)**, the equation takes the Lorentz-invariant form

$$\Box\varphi = 0.$$

This paper develops the mathematical formulation of the source-free wave equation for a massless scalar field. The four-dimensional spacetime coordinates, Minkowski metric, d'Alembertian operator, Lorentz invariance, plane-wave solutions, dispersion relation, null propagation, and massless character of the field are examined. The paper also distinguishes the mathematical wave equation from the physical interpretation assigned to the field.

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1. Introduction

The wave equation is one of the fundamental equations of mathematical physics. It occurs in many areas of classical physics, including mechanical vibrations, acoustics, electromagnetic theory, and classical field theory.

The familiar three-dimensional form is

$$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi$$

where φ represents the quantity or field undergoing propagation and c represents the characteristic propagation speed.

In mechanical systems, a wave may represent the displacement of matter or a pressure disturbance within a material medium. However, the mathematical structure of a wave equation does not by itself require that φ represent a mechanical displacement.

A classical field may possess wave dynamics without being interpreted as the displacement of a material medium.

The purpose of this paper is to formulate the classical wave equation in four-dimensional spacetime using

(x, y, z, ct)

and to examine the resulting Lorentz-invariant equation

$\Box\varphi = 0$.

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2. The Classical Wave Equation

Let

$\varphi = \varphi(x, y, z, t)$

be a scalar field.

The three-dimensional Laplacian is

$\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$.

The classical wave equation is therefore

$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi$.

Expanding the Laplacian,

$\partial^2\varphi/\partial t^2 = c^2(\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2)$.

Rearranging,

$\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2 - (1/c^2)\partial^2\varphi/\partial t^2 = 0$.

This form makes the relationship between spatial and temporal derivatives explicit.

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3. Four-Dimensional Spacetime Coordinates

We introduce the four coordinates

$x^\mu = (x, y, z, ct)$.

The scalar field can consequently be written as

$\varphi = \varphi(x, y, z, ct)$.

The use of **ct** is convenient because ct has the dimensions of length, just as x, y, and z do.

The fourth coordinate, however, is not simply another spatial coordinate. It represents time expressed in distance units.

The spacetime interval is

$$ds^2 = dx^2 + dy^2 + dz^2 - c^2dt^2$$

or

$$ds^2 = dx^2 + dy^2 + dz^2 - d(ct)^2.$$

The difference in sign between the spatial and temporal contributions is fundamental to Minkowski spacetime.

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4. Minkowski Metric

Using the metric convention adopted in this paper,

$g_{\mu\nu} = \mathrm{diag}(1, 1, 1, -1)$.

The spacetime interval can therefore be written as

$ds^2 = g_{\mu\nu}\, dx^\mu\, dx^\nu$.

The inverse metric is

$g^{\mu\nu} = \mathrm{diag}(1, 1, 1, -1)$.

The distinction between the temporal and spatial coordinates is therefore encoded directly in the metric.

This Lorentzian metric is essential for obtaining the relativistic wave operator.

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5. The Four-Dimensional Wave Operator

Define the four-dimensional derivative as

$\partial_\mu = \partial/\partial x^\mu$.

The d'Alembertian, or four-dimensional wave operator, is

$\square = g^{\mu\nu}\partial_\mu\partial_\nu$.

Using the metric above,

$\square = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2 - \partial^2/\partial(ct)^2$.

Since

$\partial/\partial(ct) = (1/c)\partial/\partial t$,

we have

$\partial^2/\partial(ct)^2 = (1/c^2)\partial^2/\partial t^2$.

Therefore,

$\square = \nabla^2 - (1/c^2)\partial^2/\partial t^2$.

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6. The Lorentz-Invariant Classical Wave Equation

The source-free scalar wave equation is

$\square\varphi = 0$.

In expanded form,

$$\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2 - \partial^2\varphi/\partial(ct)^2 = 0.$$

Using the relationship between ct and t,

$$\nabla^2\varphi - (1/c^2)\partial^2\varphi/\partial t^2 = 0.$$

Rearranging gives

$$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi.$$

Thus, the familiar classical wave equation and the four-dimensional Lorentz-invariant equation are mathematically equivalent.

The four-dimensional notation exposes the spacetime structure of the equation.

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7. Lorentz Invariance

Consider two inertial reference frames related by a Lorentz transformation:

$$x^\mu \to x'^\mu.$$

The spacetime interval remains invariant:

$$ds'^2 = ds^2.$$

For a scalar field,

$$\varphi'(x') = \varphi(x).$$

The d'Alembertian

$$\Box = g^{\mu\nu}\partial_\mu\partial_\nu$$

is constructed from the spacetime metric and therefore retains its form under Lorentz transformations.

Consequently,

$$\Box\varphi = 0$$

has the same mathematical form in every inertial reference frame.

This is the Lorentz-invariant character of the equation.

8. Plane-Wave Solution

Consider a plane-wave solution of the form

$\boldsymbol{\varphi = A\, e^{i(k\cdot r - \omega t)}}$

where

$\mathbf{r} = (x, y, z)$

and

$\mathbf{k} = (k_x, k_y, k_z)$.

For this solution,

$\nabla^2\varphi = -k^2\varphi$

and

$\partial^2\varphi/\partial t^2 = -\omega^2\varphi$.

Substitution into the wave equation gives

$-k^2\varphi + (\omega^2/c^2)\varphi = 0$.

For a nonzero field,

$\omega^2/c^2 = k^2$.

Therefore,

$\omega^2 = c^2k^2$

and

$\omega = ck$

for positive frequency.

This is the dispersion relation for the massless wave.

9. Four-Wave-Vector

The wave four-vector can be defined as

$$\mathbf{k^\mu = (k_x, k_y, k_z, \omega/c)}.$$

The plane wave may then be represented in four-dimensional notation as

$$\boldsymbol{\varphi = A\, e^{ik_u x^u}}.$$

The wave equation requires

$$\mathbf{k_\mu k^\mu = 0}.$$

With the metric used in this paper,

$$\mathbf{k_\mu k^\mu = k^2 - \omega^2/c^2}.$$

Therefore,

$$\mathbf{k_\mu k^\mu = 0}$$

gives

$$\boldsymbol{\omega^2 = c^2 k^2}.$$

The four-wave-vector is therefore a null vector.

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10. The Massless Character of the Field

The relativistic scalar-field equation for a field with mass m can be written, with the appropriate metric convention, as

$$\boldsymbol{(\Box + m^2c^2/\hbar^2)\varphi = 0}.$$

For

$$\mathbf{m = 0},$$

the mass term disappears and the equation becomes

$$\boldsymbol{\Box\varphi = 0}.$$

Thus, the source-free scalar wave equation represents the massless limit of the relativistic scalar-field equation.

The corresponding dispersion relation is

$$\omega^2 = c^2k^2.$$

The absence of a mass term is therefore directly associated with the null condition

$$k_\mu k^\mu = 0.$$

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11. Null Propagation

The Minkowski spacetime interval is

$$ds^2 = dx^2 + dy^2 + dz^2 - c^2dt^2.$$

For a null trajectory,

$$ds^2 = 0.$$

Therefore,

$$dx^2 + dy^2 + dz^2 = c^2dt^2.$$

Define the spatial distance element by

$$d\ell^2 = dx^2 + dy^2 + dz^2.$$

Then

$$d\ell = c\,dt.$$

Consequently,

$$d\ell/dt = c.$$

Thus, the characteristic propagation of the massless wave is associated with the null structure of Minkowski spacetime.

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12. Classical Field and Mechanical Wave

A mechanical wave and a classical field wave should not automatically be regarded as the same physical phenomenon.

A mechanical wave involves the dynamics of a material system. For example, a displacement field u may satisfy

$$\partial^2 u/\partial t^2 = v^2 \partial^2 u/\partial x^2.$$

In that case, u has a direct mechanical interpretation.

In contrast, the scalar field in

$$\Box\varphi = 0$$

is not required by the mathematical equation to represent a displacement of matter.

The field may instead be regarded as a classical dynamical field.

Therefore, the existence of a wave equation does not by itself establish the existence of a mechanical medium.

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13. Relation to Electromagnetic Waves

Electromagnetic radiation provides an important example of classical non-mechanical wave propagation.

In vacuum, Maxwell's equations imply wave equations for the electric and magnetic fields:

$$\Box \mathbf{E} = 0$$

and

$$\Box \mathbf{B} = 0.$$

These equations have the same d'Alembertian structure as

$$\Box\varphi = 0.$$

However, the electromagnetic field is not a scalar field. It has vector and tensor structure and is fundamentally described by the electromagnetic field tensor

$\mathbf{F_{\mu\nu}}$.

Therefore, the scalar equation

$$\Box\varphi = 0$$

should not be interpreted as a complete description of electromagnetism.

It is instead the scalar-field analogue of a massless relativistic wave equation.

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14. Lagrangian Formulation

The equation

$$\Box\varphi = 0$$

can be derived from a Lorentz-invariant classical field Lagrangian.

For a free real scalar field, consider

$$\mathcal{L} = \tfrac{1}{2}\partial_\mu\varphi\, \partial^\mu\varphi.$$

The action is

$$S = \int \mathcal{L}\, d^4x.$$

Applying the Euler-Lagrange field equation gives

$$\partial_\mu\partial^\mu\varphi = 0.$$

Therefore,

$$\Box\varphi = 0.$$

This demonstrates that the wave equation can be formulated as a classical relativistic field theory.

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15. Energy and Momentum

A classical scalar field can possess energy and momentum.

For the free scalar field, the energy-momentum tensor can be written as

$$T^{\mu\nu} = \partial^\mu\varphi\, \partial^\nu\varphi - \tfrac{1}{2}g^{\mu\nu}\partial_\alpha\varphi\, \partial^\alpha\varphi,$$

with the precise signs depending on the adopted metric and Lagrangian convention.

The field equation leads to the conservation relation

$$\partial_\mu T^{\mu\nu} = 0.$$

Thus, the classical field can carry and transport energy and momentum without requiring that φ represent a mechanical displacement.

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16. Source-Free and Source-Driven Equations

The present paper considers the source-free equation

$$\Box\varphi = 0.$$

A source can be introduced schematically through

$$\Box\varphi = J,$$

where J represents an appropriate source term.

The source-free equation describes free propagation, whereas the source equation describes a field influenced by a source.

The precise form of J depends on the physical theory under consideration.

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17. The Significance of the Four-Dimensional Form

The expression

$$\varphi = \varphi(x, y, z, ct)$$

shows that the field is defined over spacetime.

The resulting equation is

$$\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2 - \partial^2\varphi/\partial(ct)^2 = 0.$$

The temporal derivative has a sign different from the spatial derivatives.

This is not an arbitrary mathematical choice. It follows from the Lorentzian structure of Minkowski spacetime.

If all four coordinates were treated with the same sign, the resulting equation would be a Euclidean Laplace-type equation rather than a relativistic wave equation.

Therefore, the essential structure is not merely the presence of four coordinates, but the combination of four-dimensional coordinates with a Lorentzian metric.

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18. Discussion

The four-dimensional formulation provides a deeper interpretation of the classical wave equation.

The conventional expression

$$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi$$

emphasizes the relationship between temporal evolution and spatial curvature.

The covariant expression

$$\Box\varphi = 0$$

emphasizes the spacetime structure of the field.

The two forms describe the same mathematical wave equation when the Minkowski metric and the invariant propagation speed c are understood.

The formulation also clarifies that a classical wave equation does not automatically imply a mechanical medium. The physical nature of φ must be specified independently.

For a massless scalar field, the equation provides a particularly simple Lorentz-invariant model of classical field propagation.

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19. Limitations

The equation

$\Box\varphi = 0$

is not a universal equation for every physical wave.

Mechanical waves require equations appropriate to the material system in which they propagate.

Acoustic waves depend on properties of matter such as density, pressure, and compressibility.

Electromagnetic waves require Maxwell's equations.

Gravitational waves require the gravitational field equations and, in the weak-field limit, the appropriate linearized equations.

The scalar equation should therefore be interpreted specifically as the equation of a free massless scalar field.

Its significance is the general relativistic structure of wave propagation rather than a universal physical description of all wave phenomena.

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20. Conclusion

The classical wave equation

$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi$

can be naturally formulated in four-dimensional spacetime using the coordinates

$x^\mu = (x, y, z, ct)$.

With the Minkowski metric

$g_{\mu\nu} = \mathrm{diag}(1, 1, 1, -1)$,

the d'Alembertian is

$\Box = g^{\mu\nu}\partial_\mu\partial_\nu$

and explicitly,

$$\Box = \nabla^2 - (1/c^2)\partial^2/\partial t^2.$$

The source-free field equation is therefore

$$\Box\varphi = 0.$$

A plane-wave solution,

$$\varphi = A\, e^{i(k\cdot r - \omega t)},$$

produces the dispersion relation

$$\omega^2 = c^2k^2.$$

In four-vector notation, the same condition is

$$k_\mu k^\mu = 0.$$

The central mathematical result is therefore:

A free massless classical scalar field defined on Minkowski spacetime satisfies the Lorentz-invariant equation $\Box\varphi = 0$.

The four-dimensional formulation demonstrates that the classical wave equation can be understood as a field equation on spacetime. Its mathematical structure does not intrinsically require a mechanical medium. The physical interpretation of the field must, however, be specified by the particular theory in which the field is employed.

Mathematical Summary

Four-dimensional coordinates

$x^\mu = (x, y, z, ct)$

Minkowski interval

$ds^2 = dx^2 + dy^2 + dz^2 - c^2dt^2$

Metric

$g_{\mu\nu} = \mathrm{diag}(1, 1, 1, -1)$

Four-dimensional derivative

$\partial_\mu = \partial/\partial x^\mu$

Wave operator

$\Box = g^{\mu\nu}\partial_\mu\partial_\nu$

Explicit wave operator

$\Box = \nabla^2 - (1/c^2)\partial^2/\partial t^2$

Massless scalar-field equation

$\Box\varphi = 0$

Expanded equation

$\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2 - \partial^2\varphi/\partial(ct)^2 = 0$

Equivalent classical wave equation

$\partial^2\varphi/\partial t^2 = c^2\nabla^2\varphi$

Plane-wave solution

$\varphi = A\,e^{i(k\cdot r - \omega t)}$

Dispersion relation

$\omega^2 = c^2k^2$

Four-wave-vector

$k^\mu = (k_x, k_y, k_z, \omega/c)$

Null condition

$k_\mu k^\mu = 0$

Massless limit

$m = 0 \rightarrow \Box\varphi = 0$

Central equation

$\boldsymbol{\Box\varphi = 0}$